



MAROUANE BEN HADDOU

DATA ENGINEERING STUDENT

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SUMMARY

Second-year Data Engineering student at ENSAH, looking for an **end-of-year project internship (PFA)** in Data Engineering, Big Data and Machine Learning, to contribute to concrete data projects within a technical team.

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEERING INTERN — GOJI

Jul. 2026 – Sep. 2026

Paris, France (Remote)

- Contributed to the architecture and development of two AI agents: a sales agent for personalized product exploration and a hospitality concierge agent.
- Built the Goji Hospitality inbox service, enabling communication between guests and hotel staff.
- Contributed to concierge features to interpret guest requests, retrieve hotel and reservation context, and execute predefined workflows through connected systems.
- Built a CI/CD pipeline and implemented evaluations for Goji's AI sales agent.
- Collaborated with the engineering team to build the hospitality platform using Python, FastAPI, PostgreSQL, pgvector, ChromaDB, the Anthropic SDK, and the WhatsApp API.

AI ENGINEER INTERN — SMART AUTOMATION TECHNOLOGIES

Jul. 2026 – Sep. 2026

Tangier, Morocco (Remote)

- Built an AI platform coordinating specialized agents for inventory, equipment, operations and demand forecasting, with parallel execution, conflict detection, constraint validation and traceable decisions.
- Combined SQL, vector search and BM25 to ground recommendations in warehouse data and procedures.
- Developed a React/TypeScript interface and JWT-secured FastAPI backend for conversational queries and knowledge-base management; designed modular warehouse adapters with configurable policies and local LLM inference.
- Built the system using Python, LangGraph, Ollama, FastAPI, React, TypeScript, PostgreSQL, Qdrant, Redis and Docker.

DATA ANALYST INTERN — DPA (PROVINCIAL DEPARTMENT OF AGRICULTURE)

Jul. 2025 – Aug. 2025

Al Hoceima, Morocco (On-Site)

- Collection and processing of climate and agricultural data, exploratory data analysis (EDA), integration into Snowflake and Power BI dashboard for business stakeholders.
- Used Python and SQL for data processing, Snowflake for storage and Power BI for reporting.

PROJECTS

TAASIM — REAL-TIME URBAN MOBILITY PLATFORM

STREAMING / KAPPA

- Kappa architecture: 1.7M Porto trips remapped via OSRM onto Casablanca and replayed as a real-time Kafka stream.
- Three event-time Flink streaming jobs (GPS normalization, supply/demand aggregation, trip matching) with watermarks, RocksDB checkpointing to MinIO and state served from Cassandra (query-first design, TTL).
- Demand forecasting per zone and per 30-min window (Spark MLlib GBT, 413k feature rows) beating the baseline by -22% RMSE; exposed via a JWT-secured FastAPI API and Grafana dashboards (live maps, heatmaps).
- Tools:** Kafka, Flink, Spark, Cassandra, MinIO, OSRM, Grafana, FastAPI, Docker

PROCUREMENT SYSTEM BIGDATA PIPELINE

BIG DATA

- ETL pipeline orchestrated by Apache Airflow (8-task daily DAG) processing orders, inventory and master data.
- Distributed HDFS storage, Hive warehouse and Cassandra snapshots, queried through multi-source Trino federated queries.
- Automatic net demand calculation (safety stock) and automated generation of optimal supplier orders.
- Tools:** Apache Airflow, Hadoop HDFS, Apache Hive, Apache Cassandra, Trino, PostgreSQL, Docker

FINDJOB — JOB RECOMMENDATION PLATFORM

DATA ENGINEERING / AI

- Multi-source aggregation (Adzuna, ReKrute, Emploi-Public via Selenium + OCR) in a medallion architecture MongoDB and PostgreSQL (star schema).
- Hybrid recommendation engine: multilingual semantic search (pgvector + sentence-transformers) re-weighted by 7 signals (skills, seniority, contract, remote...), NLP-based CV parsing and ML salary estimation.
- Asynchronous Kafka processing (CV parsing) + Redis cache (sub-second API), React front-end and a Power BI job-market dashboard.
- Tools:** FastAPI, React, PostgreSQL/pgvector, MongoDB, Kafka, Redis, Airflow, Power BI, Docker

EDUCATION

National School of Applied Sciences (ENSAH) Al Hoceima
Engineering Degree in Data Engineering

2024 – Present

National School of Applied Sciences (ENSAH) Al Hoceima
Integrated Preparatory Cycle

2022 – 2024

Targuist Qualifying High School
Baccalaureate in Physical Sciences – French Option

2021 – 2022

ADDITIONAL INFORMATION

- Languages & Frameworks:** C, Python, Java, PL/SQL, FastAPI, React, Spring Boot
- Databases:** Oracle, MySQL, PostgreSQL, MongoDB, Snowflake, Cassandra, Redis
- Big Data & ETL:** Hadoop, Airflow, Hive, Trino, Flink, Kafka, MinIO
- Machine Learning:** Scikit-learn, Pandas, PyTorch, TensorFlow, MLflow
- Generative AI:** LangChain, LangGraph
- Scraping & Viz:** BeautifulSoup, Playwright, Power BI
- DevOps:** Git, GitHub, Docker, Grafana, Prometheus
- Methodologies & Design:** Agile/Scrum, UML, Merise
- Languages:** Arabic (Native), French (C1-TCF), English (Advanced), Spanish (Basic)